

Advanced Student Management System - Assignment

Objective

Enhance the existing Course-Student system by implementing advanced features, optimization techniques, and real-world scenarios.

1. CRUD Operations

- Edit Course
- Edit Student (including changing course)
- Delete Student
- Delete Course (handle constraints properly)

2. Advanced Filtering & Search

- Filter students by course
- Search by student name
- Search by email
- Combine filters (course + search)

3. Pagination

Implement pagination for student listing using Laravel pagination methods.

4. Aggregations

- Display total students per course
- Display total number of courses
- Find course with highest number of students

5. N+1 Problem

Demonstrate N+1 problem and fix it using eager loading (with()).

6. Form Improvements

- Show validation errors

- Retain old input values
- Keep selected course in dropdown

7. Unique Email Validation

Ensure email remains unique during update operations.

8. Course Details Page

- Show course name and duration
- List enrolled students
- Display total students
- Handle empty state (no students)

9. Edge Cases

- Course with no students
- Duplicate email handling
- Invalid course assignment

Bonus Tasks

- Implement Soft Deletes
- Restore deleted students
- Create API endpoint for students

Questions (Answer in Submission)

- Why does Student model use belongsTo?
- What is the N+1 problem?
- Why do we use with() in Eloquent?